3. cvičení

Řešení

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1. $\displaystyle\lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^{120} - \left(1+\frac{2}{n}\right)^{80}}{\left(1-\frac{1}{n}\right)^{100} + \left(1+\frac{3}{n}\right)^{100} - 2}$

Každou závorku rozvineme podle binomické věty $(1+x)^m = 1 + mx + \binom{m}{2}x^2 + \cdots + x^m$:

$$\left(1+\tfrac{1}{n}\right)^{120} = 1 + \tfrac{120}{n} + Z_n, \qquad \left(1+\tfrac{2}{n}\right)^{80} = 1 + \tfrac{160}{n} + Y_n,$$
$$\left(1-\tfrac{1}{n}\right)^{100} = 1 - \tfrac{100}{n} + G_n, \qquad \left(1+\tfrac{3}{n}\right)^{100} = 1 + \tfrac{300}{n} + H_n.$$

Jedničky a dvojka se odečtou, zůstane dominantní člen $\frac{1}{n}$:

$$\lim_{n\to\infty} \frac{-\frac{40}{n} + Z_n - Y_n}{\frac{200}{n} + G_n + H_n} = \lim_{n\to\infty} \frac{-40 + nZ_n - nY_n}{200 + nG_n + nH_n}.$$

Zbytky jsou polynomy v $\frac{1}{n}$ s nejnižší mocninou $\frac{1}{n^2}$. Zavedeme $y = \frac{1}{n}$ (Heineho věta): $y_n = \frac{1}{n} \to 0$, $y_n > 0$ a funkce $F(y) = \dfrac{(1+y)^{120} - (1+2y)^{80}}{(1-y)^{100} + (1+3y)^{100} - 2}$ je definovaná na nějakém $(0,\delta)$, protože jmenovatel je $y\,(200 + y\,q(y)) \neq 0$ pro malé $y > 0$. Zbytky jsou $Z(y) = y^2 p_1(y)$ atd., a proto

$$\lim_{y\to 0^+} \frac{-40 + y\,(p_1 - p_2)(y)}{200 + y\,(p_3 + p_4)(y)} \overset{\text{AL}}{=} \frac{-40}{200} = -\frac{1}{5}.$$

2. $\displaystyle\lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^{150} - \left(1+\frac{2}{n}\right)^{60}}{\left(1-\frac{1}{n}\right)^{99} + \left(1+\frac{3}{n}\right)^{99} - 2}$

Stejně jako v příkladu 1: čitatel je $\frac{150}{n} - \frac{120}{n} + \cdots = \frac{30}{n} + \ldots$, jmenovatel $-\frac{99}{n} + \frac{297}{n} + \cdots = \frac{198}{n} + \ldots$, zbytky jsou polynomy s nejnižší mocninou $\frac{1}{n^2}$. Po vytknutí $\frac{1}{n}$ a AL

$$\frac{30}{198} = \frac{5}{33}.$$

3. $\displaystyle\lim_{n\to\infty} \frac{\sqrt[3]{n^3+5n} - \sqrt[3]{n^3+2}}{\sqrt{n^4+n+1} - \sqrt{n^4+1}}$

Označme $A_n = \sqrt[3]{n^3+5n}$, $B_n = \sqrt[3]{n^3+2}$, $C_n = \sqrt{n^4+n+1}$, $D_n = \sqrt{n^4+1}$. Z $a - b = \dfrac{a^k - b^k}{a^{k-1} + a^{k-2}b + \cdots + b^{k-1}}$ je

$$A_n - B_n = \frac{5n-2}{A_n^2 + A_nB_n + B_n^2}, \qquad C_n - D_n = \frac{n}{C_n + D_n},$$

$$\text{výraz} = \frac{(5n-2)(C_n+D_n)}{n\,(A_n^2 + A_nB_n + B_n^2)} = \frac{\left(5-\frac{2}{n}\right)\dfrac{C_n+D_n}{n^2}}{\dfrac{A_n^2 + A_nB_n + B_n^2}{n^2}}.$$

Podle ZOAL je $\frac{A_n}{n} = \sqrt[3]{1+\frac{5}{n^2}} \to 1$, $\frac{B_n}{n} \to 1$, $\frac{C_n}{n^2} = \sqrt{1+\frac{1}{n^3}+\frac{1}{n^4}} \to 1$, $\frac{D_n}{n^2} \to 1$. Podle AL

$$\frac{5\cdot(1+1)}{1+1+1} = \frac{10}{3}.$$

4. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{n^3+n+1}-\sqrt[3]{n^3-n+2}}{\sqrt{n^4+2n+3}-\sqrt{n^4+n+4}}$

Stejný postup, $A_n = \sqrt[3]{n^3+n+1}$, $B_n = \sqrt[3]{n^3-n+2}$, $C_n = \sqrt{n^4+2n+3}$, $D_n = \sqrt{n^4+n+4}$:

$$A_n - B_n = \frac{2n-1}{A_n^2+A_nB_n+B_n^2}, \qquad C_n - D_n = \frac{n-1}{C_n+D_n},$$

$$\text{výraz} = \frac{\left(2-\frac{1}{n}\right)\dfrac{C_n+D_n}{n^2}}{\left(1-\frac{1}{n}\right)\dfrac{A_n^2+A_nB_n+B_n^2}{n^2}} \xrightarrow{\text{AL}} \frac{2\cdot 2}{1\cdot 3} = \frac{4}{3}.$$

5. $\lim\limits_{n\to\infty} \dfrac{(n+7)^{50}-(n^2+1)^{25}}{\sqrt{n^{100}+n^{99}-1}-\sqrt{n^{100}+2n^{99}+1}}$

Čitatel rozvineme binomickou větou:

$$(n+7)^{50} = n^{50}+350\,n^{49}+Z_n, \qquad (n^2+1)^{25} = n^{50}+25\,n^{48}+G_n,$$

kde Z_n, G_n jsou polynomy stupně nejvýše 48. Členy n^{50} se odečtou, takže čitatel je $n^{49}(350+r_n)$ s $r_n = \frac{Z_n - 25n^{48} - G_n}{n^{49}} \to 0$. Ve jmenovateli je

$$C_n - D_n = \frac{-n^{99}-2}{C_n+D_n}, \qquad \frac{C_n}{n^{50}} \to 1, \ \frac{D_n}{n^{50}} \to 1.$$

Proto

$$\text{výraz} = \frac{(350+r_n)\dfrac{C_n+D_n}{n^{50}}}{-\left(1+\frac{2}{n^{99}}\right)} \xrightarrow{\text{AL}} \frac{350\cdot 2}{-1} = -700.$$

6. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{n^3+6n}-\sqrt[3]{n^3+7}}{\sqrt{n^2+4}-\sqrt{n^2+1}}$

$A_n = \sqrt[3]{n^3+6n}$, $B_n = \sqrt[3]{n^3+7}$, $C_n = \sqrt{n^2+4}$, $D_n = \sqrt{n^2+1}$:

$$A_n - B_n = \frac{6n-7}{A_n^2+A_nB_n+B_n^2}, \qquad C_n - D_n = \frac{3}{C_n+D_n},$$

$$\text{výraz} = \frac{\left(6-\frac{7}{n}\right)\dfrac{C_n+D_n}{n}}{3\,\dfrac{A_n^2+A_nB_n+B_n^2}{n^2}} \xrightarrow{\text{AL}} \frac{6\cdot 2}{3\cdot 3} = \frac{4}{3}.$$

9. $\lim\limits_{n\to\infty} \dfrac{(n^2+1)^{100}-(n+2)^{200}+400n^{199}}{1+2+3+\ldots+n^{99}}$

Binomická věta: $(n^2+1)^{100} = n^{200}+100\,n^{198}+\ldots$ a
$(n+2)^{200} = n^{200}+400\,n^{199}+\binom{200}{2}\cdot 4\,n^{198}+\cdots = n^{200}+400\,n^{199}+79600\,n^{198}+\ldots$ Členy n^{200} a n^{199} se odečtou, čitatel je $-79500\,n^{198}+(\text{nižší mocniny})$. Jmenovatel je $\dfrac{n^{99}(n^{99}+1)}{2} = \dfrac{n^{198}}{2}\left(1+\frac{1}{n^{99}}\right)$. Po vydělení n^{198} a AL

$$\frac{-79500}{\frac{1}{2}} = -159000.$$